



Basic Introduction to Gene Expression Analysis: Key Concepts (Part 1)

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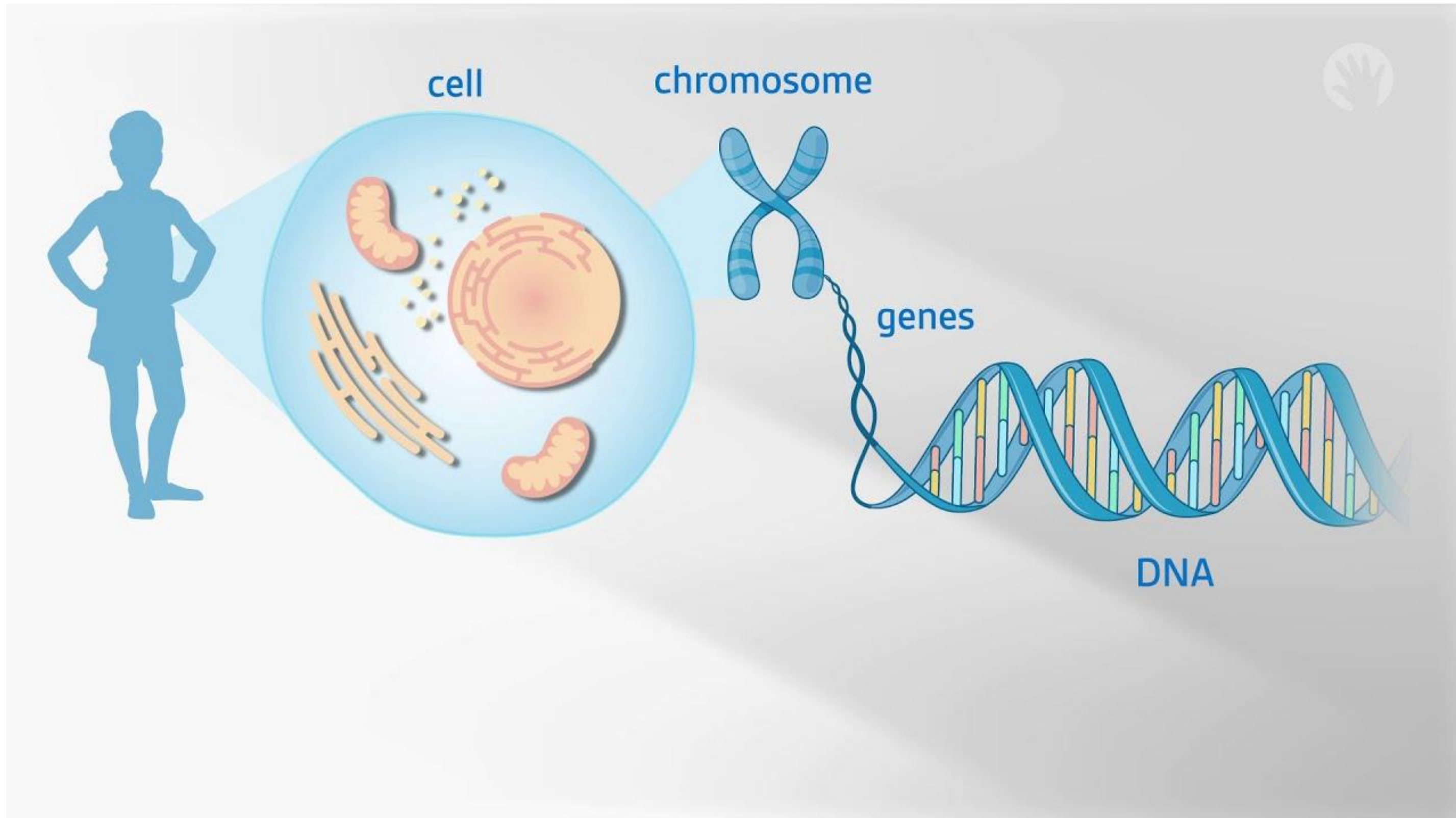
Learning Objectives

- **Understand the central dogma** of molecular biology and its relationship to gene expression.
- **Describe the importance of gene regulation** and how it affects biological function and disease.
- **State why gene expression analysis is important** in computational biology and biomedical research.

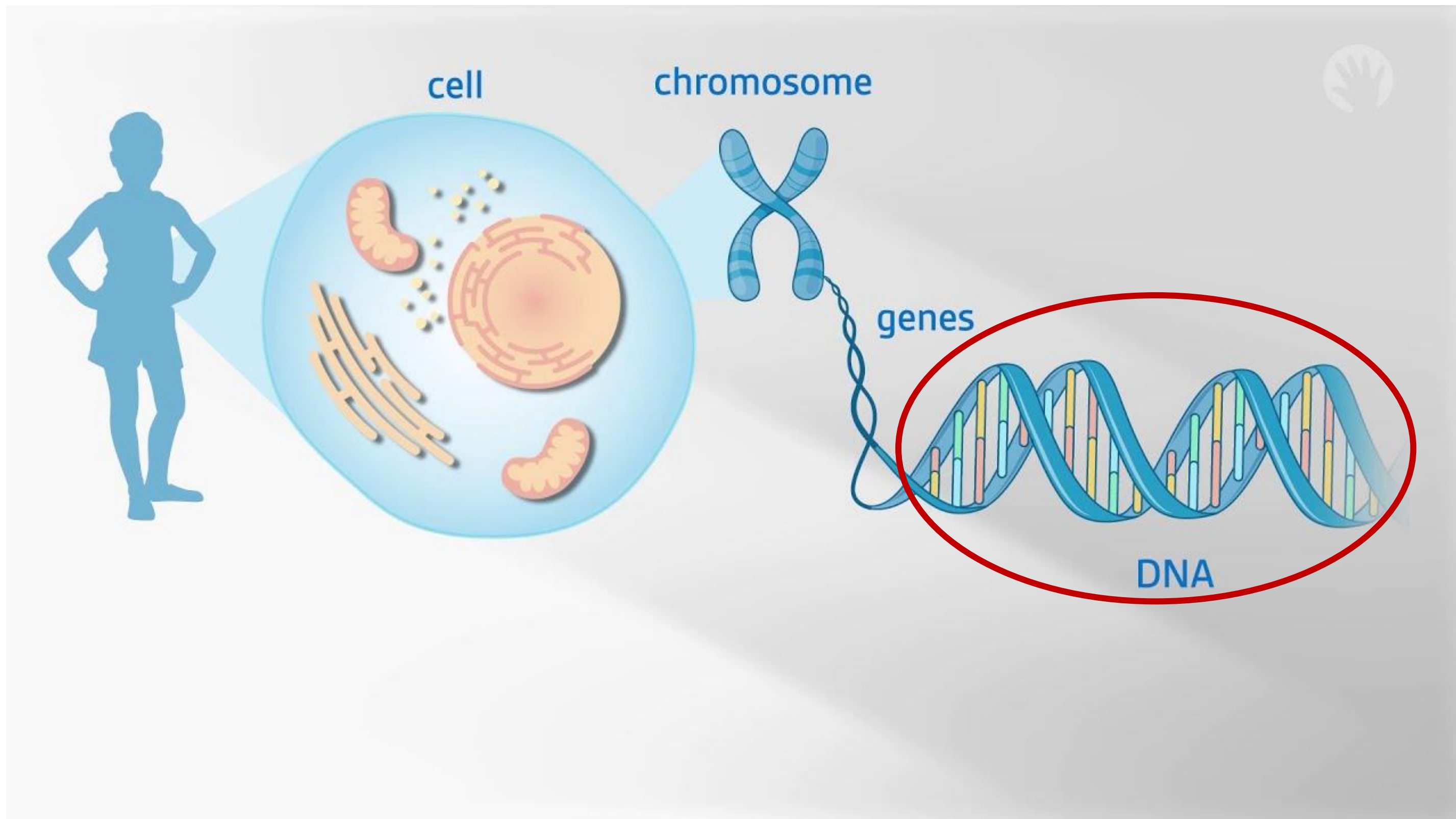
Learning Objectives

- **List major databases** that store gene expression data (e.g., GEO, ArrayExpress, GTEx, TCGA).
- **Recognize key tools and software** used in gene expression analysis workflows (e.g., DESeq2, limma, edgeR).

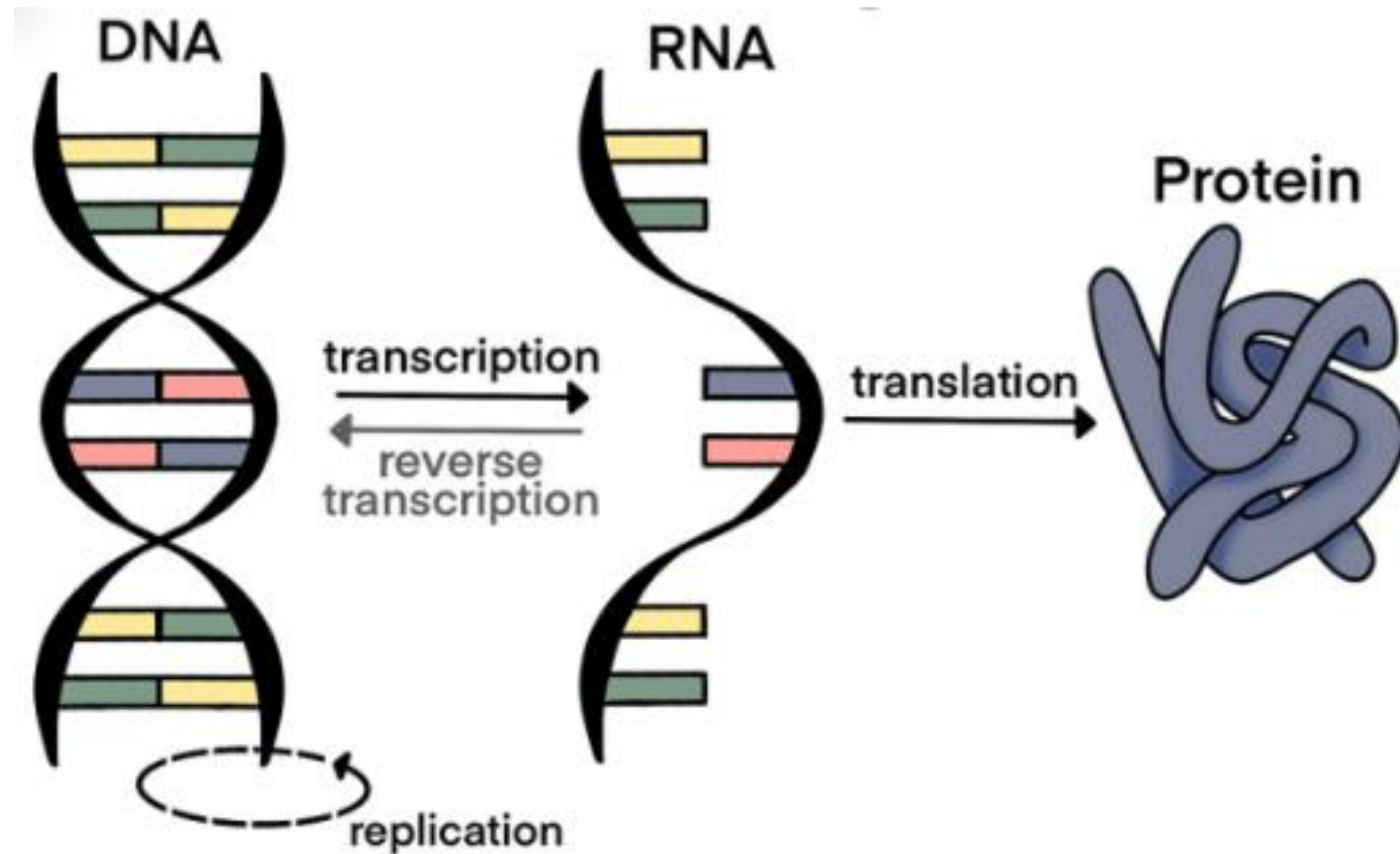
RECAP: Molecular Biology



RECAP: Molecular Biology



RECAP: The Central Dogma



What does it mean for a
gene to be **expressed**?

Gene expression

- Gene expression is how **genetic information** stored in DNA is **converted** into functional molecules like **proteins**.
- It determines **cell function, structure** and **response** to the environment.

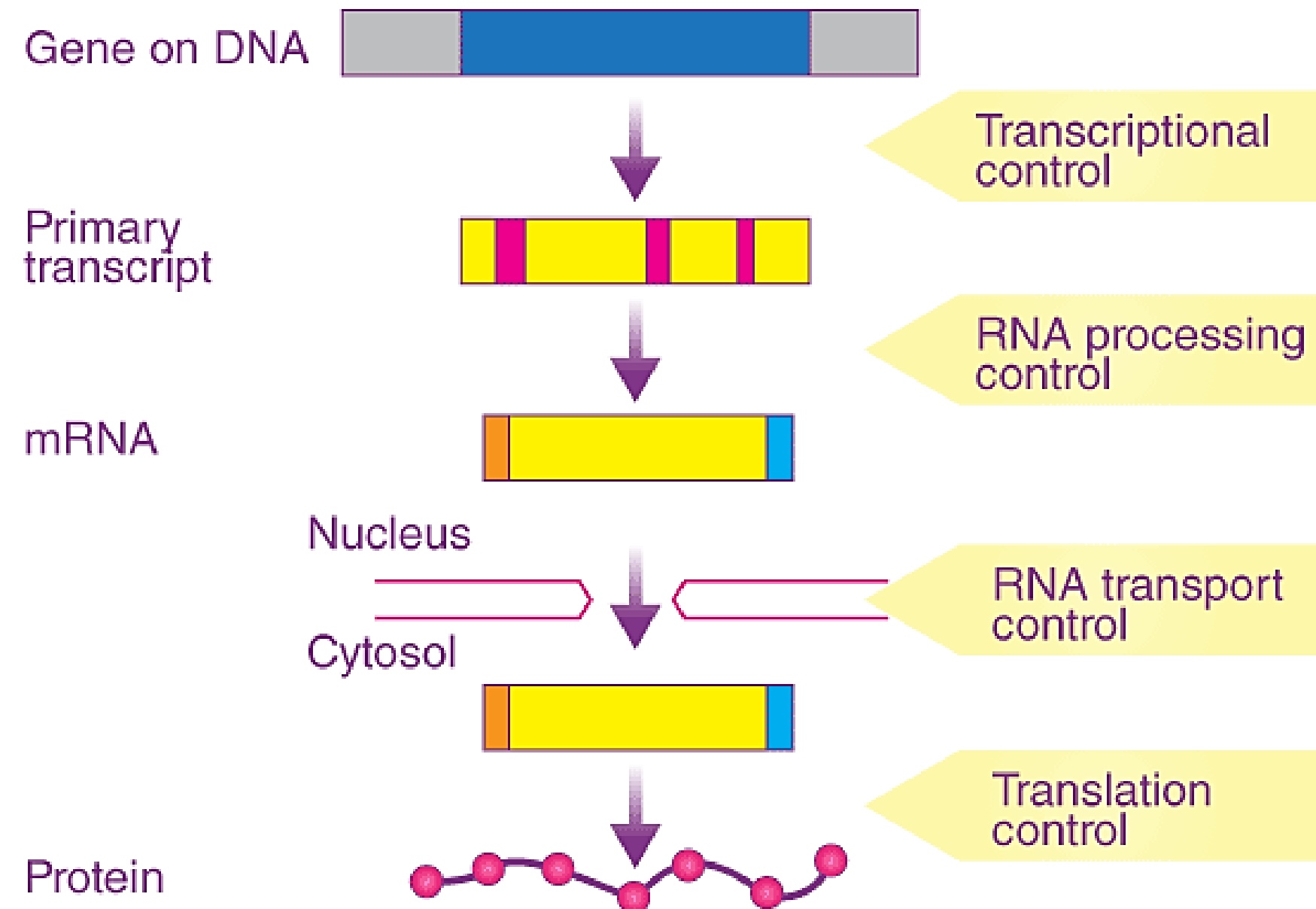
Do you know?

A liver cell and a brain cell have the same DNA, but they express different genes.

Gene Regulation

- Gene expression is controlled at multiple levels to ensure the right proteins are made at the right time.
- **Controls timing, location and amount of gene expression.**

Gene Regulation



Key Concepts

- **Upregulation:** A gene is expressed **more**, producing more mRNA and protein.
- **Downregulation:** A gene is expressed **less**, producing less mRNA and protein.
- **Example:** cortisol, ghrelin, insulin, growth hormones.

Factors

- **Environmental:** Changes in temperature, food availability, stress, etc.
- **Developmental stages:** Genes active in childhood may be inactive in adults.
- **Cell Type-Specific Needs:** A liver cell doesn't need the same proteins as a brain cell.

What are non-expressed genes called?

**Why is gene expression analysis
important?**

Importance of Gene expression

- Gene expression drives **cell function, disease mechanisms** and **drug responses**.
- Gene expression analysis reveals how genes function in different conditions.
- **Applications:** Disease research, Drug Discovery, Biomarker Identification, Agriculture and Biotechnology.

Differential gene expression (DGE) Analysis

- Comparing gene expression levels between conditions (e.g., diseased vs. healthy).
- Helps identify **biomarkers** and **drug targets**.

Key Concepts

- **Differentially expressed genes (DEGs):** are genes that show significant differences in expression levels between two or more conditions.
- **Example:** In cancer, **oncogenes** (cancer-promoting genes) might be highly expressed, while **tumor suppressor genes** might be downregulated.

Key Concepts

- **Log2 Fold Change (log2FC):** Measures the magnitude of expression difference between conditions.

$$\log_2 FC = \log_2 \left(\frac{\text{Expression in Condition B}}{\text{Expression in Condition A}} \right)$$

Key Concepts

- **P-values:** assess the **statistical significance** of observed differences.
- **Adjusted P-values (False Discovery Rate, FDR):** FDR correction (like Benjamini-Hochberg) is applied to reduce false positives.

Key Concepts

- **Normalisation:** Adjusts for differences in **sequencing depth** or **background noise** so that samples are comparable.
- **Batch effects:** Technical variations that **affect** gene expression patterns.

Experimental approaches

- **Microarrays:** Uses labeled probes to detect known gene sequences.
- **RNA-seq:** Uses **next-generation sequencing (NGS)** to read all RNA molecules in a sample.

Experimental approaches

- **Single-cell RNA-Seq (scRNA-Seq):** Captures gene expression at a single-cell level, useful for identifying rare cell types..
- **qPCR (Quantitative PCR):** Used to validate a few genes, not for whole transcriptome analysis.

Databases for Gene Expression Data

**1. The Cancer
Genome
Atlas (TCGA)**

2. ArrayExpress

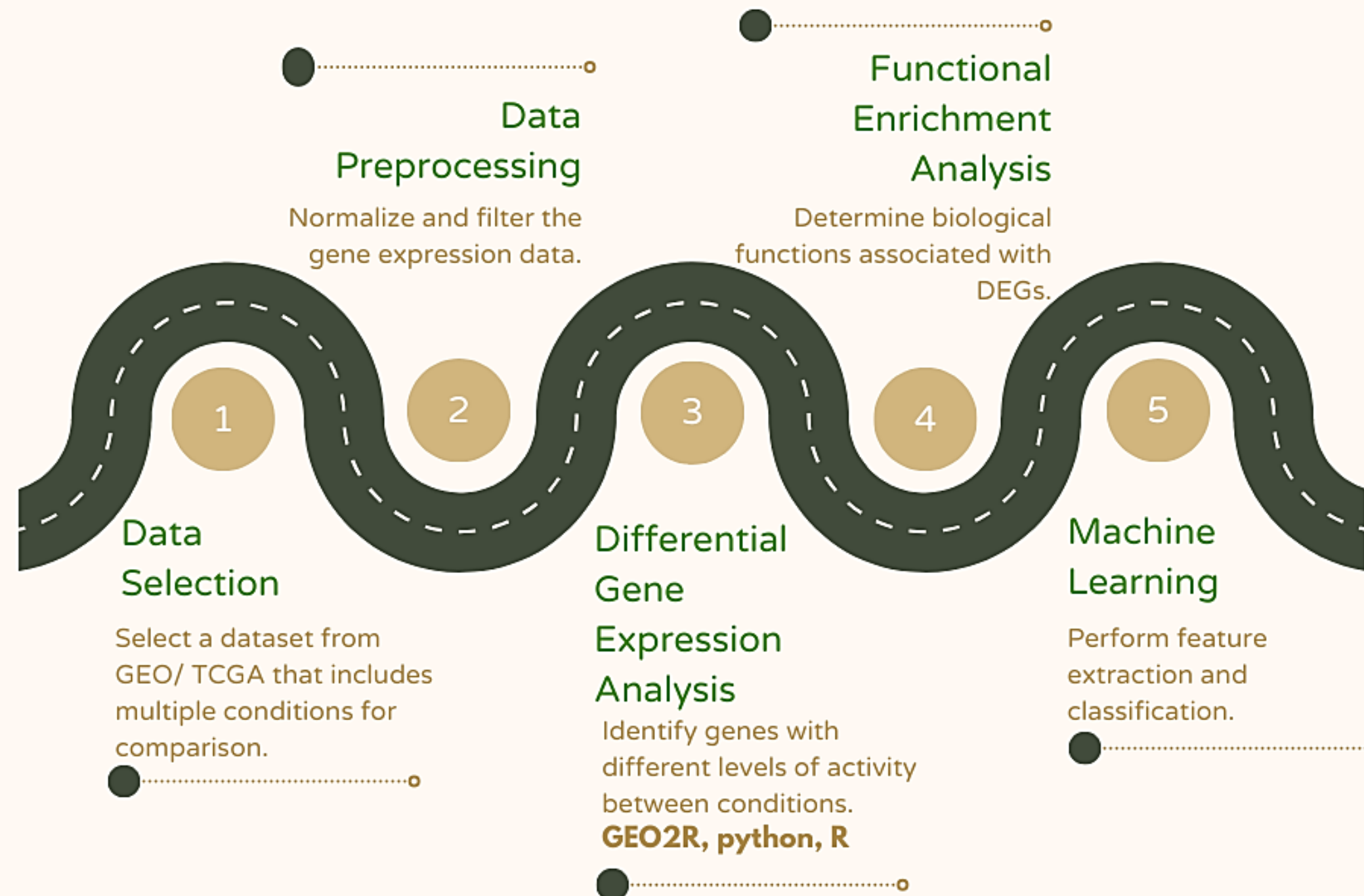
**3. Gene Tissue
Expression
(GTEx)**

**4. Gene Expression
Omnibus (GEO)**

Tools for Gene Expression Analysis

- **GEO2R:** Online tool for analyzing GEO datasets (microarray & RNA-Seq).
- **Galaxy:** for bioinformatics workflows, including RNA-Seq and microarray analysis
- **iDEP:** Interactive tool for RNA-Seq and microarray data analysis, including normalisation and pathway analysis.

Gene Expression Analysis Workflow



Roadmap by:
Chioma Onyido

Conclusion

- Before we move into real data, it's important you understand the *why* behind **gene expression analysis**.

These core concepts: **gene regulation, tools and databases** will help you ask better questions and make more sense of the results when we start analysing expression data in the next session.

Further Reading/ References

- **Bioconductor (2025).** Analyzing RNA-seq data with DESeq2
<https://bioconductor.org/packages/devel/bioc/vignettes/DESeq2/inst/doc/DESeq2.html>
- **Ann and Robert.** How do Genes work?
<https://www.luriechildrens.org/en/specialties-conditions/pediatric-genetic-disorders/about-genes/>
- **BYJU's.** Gene Regulation. <https://byjus.com/biology/gene-regulation/>
- **Nature Education.** Gene Expression.
<https://www.nature.com/scitable/topicpage/gene-expression-14121669/>

Coming up next...

Differential Gene Expression Analysis:

- ✓ Practical session:
- ✓ RNA-Seq data
- ✓ Normalisation
- ✓ Visualisation